SIMULTANEOUS CONFIDENCE BANDS FOR FUNCTIONAL VARYING-COEFFICIENT MODELS

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Abstract: We develop quantitative simultaneous inference for functional varying-coefficient models under sparse and dense sampling. A subject-level linear representation connects local polynomial estimation to high-dimensional Gaussian approximation without reconstructing individual trajectories. Explicit error bounds account for dimension, sampling frequency, smoothing bias, estimated designs, and fitted residuals. A Gaussian multiplier bootstrap calibrates joint confidence bands over time and coefficients. The theory allows growing regression dimension and extends to high-dimensional functional means. Simulations and a liver-disease biomarker analysis illustrate the procedure.

Key words and phrases: Functional data, Gaussian approximation, multiplier bootstrap, simultaneous inference, varying-coefficient model.

1. Introduction

Simultaneous inference for functional data requires uncertainty to be controlled over an entire time interval and, for multivariate targets, across several functions at once. This problem is particularly challenging under sparse and irregular sampling, when individual trajectories cannot be accurately reconstructed from their own measurements. Varying-coefficient models (VCMs) provide an important setting: they describe associations that evolve over time through a vector of coefficient functions (Hastie and Tibshirani, 1993; Gu et al., 2014; Cao, Liu and Zhou, 2018). Joint confidence bands must account for dependence within subjects and across coefficient estimates, while remaining valid as the number of observations per subject increases. We study this problem through quantitative Gaussian and bootstrap approximation, with explicit control of the errors that determine simultaneous coverage.

The sampling design has long been central to functional data analysis; see Wang, Chiou and Müller (2016). Dense-data inference can exploit accurate trajectory reconstruction (Zhang and Chen, 2007), whereas sparse-data methods pool information across subjects (Yao, Müller and Wang, 2005). Uniform estimation theory and optimal mean-estimation rates clarify the transition between these settings (Li and Hsing, 2010; Cai and Yuan,

2011). Zhang and Wang (2016) also distinguish equal-subject and equal-observation weighting when counts differ. These choices change how sampling frequency enters the variance and cannot be interchanged without checking the resulting theory.

Simultaneous inference is available in both classical regimes. For mean curves, dense-data bands include Degras (2011); Cao, Yang and Todem (2012), and sparse-data bands include Ma, Yang and Carroll (2012); Zheng, Yang and Härdle (2014). For regression, local polynomial and spline methods provide established estimation and inference tools (Wu and Chiang, 2000; Huang, Wu and Zhou, 2004; Fan and Zhang, 2000). Gu et al. (2014) develop sparse-data coefficient bands with time-independent covariates, while Cao, Liu and Zhou (2018) allow time-varying covariates under sparse irregular sampling. Dense functional-response regression is studied by Zhu, Li and Kong (2012); Chang, Lin and Ogden (2017). Zhou et al. (2024) develop empirical-likelihood M-estimation and simultaneous regions for functional responses with subject-level covariates observed on a common grid. More recently, Chang and McKeague (2026) construct empirical-likelihood bands for concurrent regression on a dense fixed grid, allowing nonsmooth functional features. These dense-grid results and sparse random-design results address different observation schemes.

Several inferential methods accommodate a range of observation frequencies. Kim and Zhao (2013) study longitudinal mean inference, Chen and Yao (2017) develop self-normalized pointwise inference for varying coefficients, and Wang et al. (2018) provide empirical-likelihood pointwise regions and global tests for concurrent functional regression. These pointwise and global procedures address different inferential targets from bands calibrated jointly over time and coefficient functions.

For scalar mean functions, Cai and Hu (2025) and Lou and Wu (2026) develop unified simultaneous inference using B-spline and local linear estimators, respectively. Their Gaussian approximations yield asymptotically valid bands and characterize covariance transitions across sampling frequencies. Our focus is quantitative simultaneous inference for vector-valued functional parameters throughout this transition.

We take a nonasymptotic approach based on high-dimensional Gaussian approximation and comparison theory (Chernozhukov, Chetverikov and Kato, 2017; Chernozhukov et al., 2022). Representing the estimation error through independent subject contributions allows us to analyze sparse observations without reconstructing individual trajectories. We derive explicit bounds for Gaussian approximation, conditional bootstrap approximation, and joint coverage error. These bounds quantify how sample size,

dimension, observation frequency, and bandwidth determine inferential accuracy, providing a quantitative explanation of the sparse-to-dense phase transition and distinguishing its consequences for approximation error and band width.

A Gaussian multiplier bootstrap calibrates the joint maximum directly from fitted subject contributions, avoiding separate estimation of the functional-error covariance surface or selection of a sampling regime. We establish its validity while accounting for fitted residuals, estimated design matrices, and studentization. The framework accommodates local polynomial estimators of any fixed degree and yields joint bands for VCMs with random time-varying predictors and growing dimension. We further extend the analysis to high-dimensional functional means, with simultaneous coverage over time and a growing collection of coordinates.

Section 2 introduces the model, estimator, and regularity conditions. Section 3 develops the linear representation, Gaussian and bootstrap approximation, simultaneous bands, and a high-dimensional mean extension. Section 4 reports simulations and an exploratory Mayo PBC analysis. The Supplementary Material provides proofs, additional sampling-frequency and mean results, and bandwidth and implementation details.

2. Model and regularity conditions

2.1 Model and estimation

Let $Y_i(t)$ be a scalar response and $\mathbf{X}_i(t) = \{X_{i1}(t), \ldots, X_{ip}(t)\}^\intercal$ a vector of predictors for subject i at time $t \in \mathcal{T} = [0, 1]$. We consider the functional varying-coefficient model

$$Y_i(t) = \mathbf{X}_i(t)^\intercal \boldsymbol{\beta}(t) + \varepsilon_i(t), \qquad i = 1, \ldots, n, \tag{2.1}$$

where $\boldsymbol{\beta}(t) = \{\beta_1(t), \ldots, \beta_p(t)\}^\intercal$ is a vector of unknown coefficient functions and $\varepsilon_i(t)$ is a functional error process. The coefficient $\beta_k(t)$ describes the linear association between the response and the kth predictor at time t, with the other predictors included in the model. A constant coordinate $X_{i1}(t) \equiv 1$ supplies an intercept; time-independent covariates can likewise be included as constant coordinate processes. Both the predictors and their associations with the response may vary over time, and the dimension $p = p_n$ may increase with n.

The subject processes are independent across i, while dependence within a subject is retained. We impose the pointwise orthogonality condition $\mathbb{E}\{\mathbf{X}_i(t)\varepsilon_i(t)\} = \mathbf{0}$. Writing $\boldsymbol{\Omega}(t) = \mathbb{E}\{\mathbf{X}_i(t)\mathbf{X}_i(t)^\intercal\}$, this condition identifies the population linear regression coefficient as

$$\boldsymbol{\beta}(t) = \boldsymbol{\Omega}(t)^{-1}\mathbb{E}\{\mathbf{X}_i(t)Y_i(t)\}$$

whenever $\mathbf{\Omega}(t)$ is nonsingular. In particular, this formulation does not require the predictor and error processes to be independent.

The trajectories are observed at subject-specific times. Subject i supplies $(T_{ij}, \mathbf{X}_{ij}, Y_{ij})$, $j = 1, \ldots, m_i$, where

$$\mathbf{X}_{ij} = \mathbf{X}_i(T_{ij}), \qquad Y_{ij} = Y_i(T_{ij}), \qquad \varepsilon_{ij} = \varepsilon_i(T_{ij}).$$

The observation times have a common density f and are independent of the subject processes. The counts m_i may differ across subjects and may remain bounded or increase with n. We pool observations across subjects for estimation and simultaneous inference over all $t \in \mathcal{T}$ and all p coefficient functions, without first reconstructing individual trajectories.

To construct a local polynomial estimator, fix a degree $r \geq 0$ and a bandwidth $h > 0$. At every target time $t \in \mathcal{T}$, approximate each coefficient function by its Taylor polynomial in a neighborhood of t. Under the smoothness condition in Assumption 2,

$$\beta_k(v) = \sum_{a=0}^{r} \frac{h^a \beta_k^{(a)}(t)}{a!} \left(\frac{v-t}{h} \right)^a + O(|v-t|^{r+1}), \qquad |v-t| \leq h,$$

where $v \in \mathcal{T}$, $\beta_k^{(0)} = \beta_k$, and the remainder bound is uniform in k and t. Thus the coefficient of the normalized power $\{(v-t)/h\}^a$ approximates $h^a \boldsymbol{\beta}^{(a)}(t)/a!$. The polynomial approximation concerns the coefficient functions; the predictors enter through their observed values $\mathbf{X}_{ij}$.

Let K be a nonnegative kernel supported on $[-1,1]$ and put $K_h(u) = K(u/h)/h$. Define

$$\boldsymbol{\theta}_r(u) = (1, u, \ldots, u^r)^\top, \qquad \boldsymbol{\psi}_{ij}(t) = \boldsymbol{\theta}_r\{(T_{ij} - t)/h\} \otimes \mathbf{X}_{ij},$$

where $\otimes$ denotes the Kronecker product. Write the local polynomial coefficients as a block vector

$$\mathbf{b} = (\mathbf{b}_0^\top, \ldots, \mathbf{b}_r^\top)^\top \in \mathbb{R}^{(r+1)p}.$$

The corresponding local predictor is

$$\boldsymbol{\psi}_{ij}(t)^\top \mathbf{b} = \sum_{a=0}^{r} \mathbf{X}_{ij}^\top \mathbf{b}_a \left(\frac{T_{ij} - t}{h}\right)^a.$$

We estimate the blocks by equal-subject weighted least squares:

$$\widehat{\mathbf{b}}(t) = \arg\min_{\mathbf{b} \in \mathbb{R}^{(r+1)p}} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t)\{Y_{ij} - \boldsymbol{\psi}_{ij}(t)^\top \mathbf{b}\}^2. \tag{2.2}$$

The kernel localizes the fit to observations within distance h of t. The factor $1/m_i$ first averages the losses within each subject, so that every subject receives the same total weight before localization. This weighting keeps subjects with more observations from receiving greater total weight solely because of their observation counts.

For computation, define the empirical and population design matrices

$$\widehat{\mathbf{D}}_h(t) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t) \boldsymbol{\psi}_{ij}(t) \boldsymbol{\psi}_{ij}(t)^\top, \qquad \mathbf{D}_h(t) = \mathbb{E}\{\widehat{\mathbf{D}}_h(t)\}.$$

When $\widehat{\mathbf{D}}_h(t)$ is invertible, the normal equations for (2.2) give

$$\widehat{\mathbf{b}}(t) = \widehat{\mathbf{D}}_h(t)^{-1}\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}K_h(T_{ij}-t)\boldsymbol{\psi}_{ij}(t)Y_{ij}.$$

Partition $\widehat{\mathbf{b}}(t)$ into its p-dimensional blocks $\widehat{\mathbf{b}}_0(t),\ldots,\widehat{\mathbf{b}}_r(t)$. Evaluating the fitted polynomial at its center gives the coefficient estimator

$$\widehat{\boldsymbol{\beta}}(t) = \widehat{\mathbf{b}}_0(t) = \mathbf{L}\widehat{\mathbf{b}}(t), \qquad \mathbf{L} = [\mathbf{I}_p\ \mathbf{0}\ \cdots\ \mathbf{0}].$$

The same fit also supplies derivative estimators. Differentiating the fitted polynomial with respect to its time argument and evaluating at t yields

$$\widehat{\boldsymbol{\beta}}^{(a)}(t) = \frac{a!}{h^a}\widehat{\mathbf{b}}_a(t), \qquad a = 0,\ldots,r, \tag{2.3}$$

with $\widehat{\boldsymbol{\beta}}^{(0)} = \widehat{\boldsymbol{\beta}}$. For local linear fitting ($r = 1$), the first two blocks estimate $\boldsymbol{\beta}(t)$ and $h\boldsymbol{\beta}'(t)$, respectively, so that $\widehat{\boldsymbol{\beta}}'(t) = \widehat{\mathbf{b}}_1(t)/h$. For $r \geq 2$, the second derivative is estimated by $2\widehat{\mathbf{b}}_2(t)/h^2$. The notation in (2.3) refers to derivatives of the local fitted polynomial at its center; these estimators need not equal derivatives of the curve obtained by moving the fitting center t. The simultaneous bands developed below concern the coefficient functions corresponding to $a = 0$.

For a vector or matrix function A, $\|A\|_{\infty,I}$ is the largest absolute entry, maximized over $t \in I$; $\|A\|_{2,I}$ is the analogous supremum of its Euclidean norm when A is a vector. Constants in $\lesssim$ and O_P are independent of n, p, h

and the counts. We use $\mathcal{F}_n$ for the sigma-field generated by all observations; conditional probabilities and expectations are written explicitly given $\mathcal{F}_n$.

2.2 Regularity conditions

The dimension $p = p_n$ may grow; r and $q \geq 4$ are fixed. All constants in the following conditions are uniform in n, p, and the observation counts. We use $\|\cdot\|_{\mathrm{op}}$ for the spectral norm and $\preceq$ for the positive-semidefinite order. All processes are assumed jointly measurable and separable, and all suprema below are measurable. All time-indexed conditions hold uniformly over $\mathcal{T} = [0, 1]$.

Assumption 1 (Sampling)**.** *For each n, the pairs $(\mathbf{X}_i, \varepsilon_i)$ are independent and identically distributed across subjects, and the counts $m_i \geq 1$ are deterministic. Their common process law may depend on n. The times T_{ij} are mutually independent with common density f, and independent of the subject processes. The model* (2.1) *satisfies $\mathbb{E}\{\mathbf{X}(t)\varepsilon(t)\} = \mathbf{0}$ for all t.*

Assumption 2 (Smoothness and identification)**.** *The coordinates of $\boldsymbol{\beta}$ belong to a common bounded $C^{r,1}(\mathcal{T})$ class: they have r continuous derivatives, and their rth derivatives are Lipschitz; for $r = 0$, the functions themselves are Lipschitz. The density f is Lipschitz and $0 < c_f \leq f \leq C_f$. With*

$\mathbf{\Omega}(t) = \mathbb{E}\{\mathbf{X}(t)\mathbf{X}(t)^\intercal\}$,

$$c_\Omega \mathbf{I}_p \preceq \mathbf{\Omega}(t) \preceq C_\Omega \mathbf{I}_p, \qquad \|\mathbf{\Omega}(t) - \mathbf{\Omega}(u)\|_{\mathrm{op}} \le C|t-u|.$$

Assumption 3 (Kernel and bandwidth)**.** *The kernel K is nonnegative, symmetric, Lipschitz on $\mathbb{R}$, supported on $[-1,1]$, and has integral one. The bandwidth is deterministic, $0 < h \le 1/2$, and $h \to 0$.*

Assumption 4 (Coordinate moments)**.** *For a fixed $q \ge 4$,*

$$\max_{k\le p} \mathbb{E}\{\sup_t |X_k(t)|^{2q}\} + \max_{k\le p} \mathbb{E}\{\sup_t |X_k(t)\varepsilon(t)|^q\} \le C.$$

Write

$$F_i = 1 + \sup_t \|\mathbf{X}_i(t)\|_\infty^2 + \sup_t \|\mathbf{X}_i(t)\varepsilon_i(t)\|_\infty, \qquad \|F_i\|_{L^q} \le K_n.$$

The coordinate moment bounds yield the worst-case upper bound $\|F_i\|_{L^q} \le Cp^{1/q}$, and we can take $1 \le K_n \le Cp^{1/q}$. Stronger assumptions that yield smaller K_n lead to sharper rates.

Assumption 5 (Score covariance)**.** *Define the score covariance kernel by*

$$\mathbf{C}(u,v) = \mathbb{E}\{\mathbf{X}(u)\varepsilon(u)\varepsilon(v)\mathbf{X}(v)^\intercal\}.$$

It satisfies

$$\sup_{u,v} \|\mathbf{C}(u,v)\|_{\mathrm{op}} \le C,$$

$$\|\mathbf{C}(u,v) - \mathbf{C}(u',v')\|_{\mathrm{op}} \le C(|u-u'| + |v-v'|).$$

For $\mathbf{V}_F(t) = \mathbf{\Omega}(t)^{-1}\mathbf{C}(t,t)\mathbf{\Omega}(t)^{-1}$, $\inf_{t\in\mathcal{T}} \min_{k\le p} [\mathbf{V}_F(t)]_{kk} \ge c_F > 0$.

Assumptions 1–3 describe independently sampled subjects with noninformative random observation times, smooth coefficients, and a uniformly well-conditioned design, while retaining within-subject dependence. Examples include well-conditioned banded second-moment matrices with uniformly Lipschitz entries and a fixed number of nonzero diagonals, and the Epanechnikov kernel used in our numerical studies.

Assumptions 4–5 accommodate growing dimension, polynomial tails of sufficiently high order, and dependence across times and coordinates, with K_n recording the size of the largest-coordinate envelope. The coordinate variance condition holds, for example, for mean-zero functional errors independent of the predictors with uniformly positive variances. The general condition also allows singular score covariance matrices.

3. Simultaneous inference

The argument follows three steps: a subject-level linear representation, Gaussian and bootstrap approximation, and calibration of the joint bands.

3.1 Linear representation

Define the subject score and its population-weighted contribution by

$$\mathbf{u}_i(t) = \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t) \boldsymbol{\psi}_{ij}(t) \varepsilon_{ij}, \qquad \boldsymbol{\Psi}_i(t) = \mathbf{L}\mathbf{D}_h(t)^{-1} \mathbf{u}_i(t).$$

The $\mathbf{\Psi}_i$ are independent and centered. Put $s = r + 1$ and

$$\kappa_{a,n} = \frac{1}{n}\sum_{i=1}^{n} m_i^{-a} \quad (a = 1, 3), \qquad v_n^2 = 1 + \frac{\kappa_{1,n}}{h}, \qquad L_n = \log(np/h),$$
$$\delta_n = v_n\sqrt{\frac{L_n}{n}} + \frac{K_n L_n}{n^{1-1/q}h},$$
$$\rho_n = p^{q/(2q-2)} v_n\sqrt{\frac{L_n}{n}} + \frac{pL_n}{n^{1-1/q}h}. \tag{3.1}$$

Here $K_n \geq 1$ bounds the L^q norm of the largest coordinate design and score envelope, defined in Assumption 4. The scale v_n^2 is the order of the average coordinate variance of the subject contributions. The rate δ_n controls score coordinates, whereas ρ_n controls the empirical design in spectral norm. Direct matrix moment bounds give $\rho_n \lesssim p\delta_n$. The matrix envelope has L^q norm of order at most p, directly from the coordinate moments.

Theorem 1 (Uniform linear representation)**.** *Under Assumptions 1–4, if $\rho_n \to 0$, the empirical design is uniformly invertible on $\mathcal{T}$ with probability tending to one, and*

$$\widehat{\boldsymbol{\beta}}(t) - \boldsymbol{\beta}(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{\Psi}_i(t) + \mathbf{R}_n(t), \qquad t \in \mathcal{T},$$

where

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_{2,\mathcal{T}} = O_P\{\sqrt{p}(\delta_n + h^s)\},$$
$$\|\mathbf{R}_n\|_{\infty,\mathcal{T}} = O_P\{\sqrt{p}(\rho_n\delta_n + h^s)\}.$$

The remainder records both estimation of the local design and smoothing bias. Its bound over the full fitting interval also controls residuals evaluated at the observed times. When the variance terms dominate ρ_n and δ_n, the stochastic remainder has order $p^{1+1/(2q-2)} v_n^2 L_n/n$. The factor $\sqrt{p}$ in the bias bound reflects coordinatewise smoothness of the p coefficient functions.

3.2 Gaussian and bootstrap approximation

Let $\tau_k^2(t) = n^{-1} \sum_i \mathbb{E}\{\Psi_{ik}^2(t)\}$ and let G_n be a centered continuous Gaussian process with

$$\mathbb{E}\{G_{n,k}(u) G_{n,l}(t)\} = \frac{\sum_{i=1}^n \mathbb{E}\{\Psi_{ik}(u)\Psi_{il}(t)\}}{n\tau_k(u)\tau_l(t)}.$$

Under Assumption 5, $\tau_k(t) \asymp v_n$ uniformly; the covariance calculation is in Section S1.2. For comparison with $S_{G,n} = \|G_n\|_{\infty,\mathcal{T}}$, define

$$T_n^\circ = \sup_{t\in\mathcal{T}} \max_{k\le p} \frac{\sqrt{n}\,|\widehat{\beta}_k(t) - \beta_k(t)|}{\tau_k(t)}.$$

Set $T_n^\circ = 0$ on the exceptional design event. Define the moment scales and the Gaussian approximation error by

$$\begin{aligned} \Lambda_n &= \frac{1 + \kappa_{3,n}/h^3}{v_n^4}, \qquad B_n = \max\left\{ \frac{\sqrt{p}\,K_n}{h v_n}, p\sqrt{\Lambda_n} \right\}, \\ \gamma_n &= \left(\frac{B_n^2 L_n^5}{n} \right)^{1/4} + \frac{B_n L_n^{3/2-1/q}}{n^{1/2-1/q}} \\ &\quad + \left(\frac{\sqrt{npL_n}\,\rho_n\delta_n}{v_n} \right)^{1/2} + \frac{\sqrt{npL_n}\,h^s}{v_n}. \end{aligned} \tag{3.2}$$

Theorem 2 (Gaussian approximation)**.** *Under Assumptions 1–5, if* $\rho_n \to 0$*, then, for all sufficiently large* n*,*

$$\sup_z |\mathbb{P}(T_n^\circ \le z) - \mathbb{P}(S_{G,n} \le z)| \le C\gamma_n.$$

The first two terms of γ_n control Gaussian approximation of the subject scores; the remaining terms account for design estimation and smoothing bias. Smaller discretization and exceptional-event terms are absorbed into this bound, as shown in the proof.

For implementation, form $\widehat{\varepsilon}_{ij} = Y_{ij} - \mathbf{X}_{ij}^\intercal \widehat{\boldsymbol{\beta}}(T_{ij})$ and

$$\begin{aligned} \widehat{\boldsymbol{\Psi}}_i(t) &= \mathbf{L}\widehat{\mathbf{D}}_h(t)^{-1} \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t)\boldsymbol{\psi}_{ij}(t)\widehat{\varepsilon}_{ij}, \\ \widehat{\sigma}_k^2(t) &= \frac{1}{n}\sum_{i=1}^{n} \widehat{\Psi}_{ik}^2(t). \end{aligned} \tag{3.3}$$

Independently of the data, draw independent $\xi_i \sim N(0,1)$ and set

$$G_{n,k}^*(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \xi_i \frac{\widehat{\Psi}_{ik}(t)}{\widehat{\sigma}_k(t)}, \qquad S_n^* = \|G_n^*\|_{\infty,\mathcal{T}}. \tag{3.4}$$

A single multiplier is shared by all observations and coordinates of a subject. The statistic being calibrated is

$$T_n = \sup_{t\in\mathcal{T}} \max_{k\le p} \frac{\sqrt{n}\,|\widehat{\beta}_k(t) - \beta_k(t)|}{\widehat{\sigma}_k(t)}.$$

Collect the covariance and fitted-score errors in

$$e_n = p\left\{ \sqrt{\frac{\Lambda_n L_n}{n}} + \frac{K_n^2 L_n}{n^{1-2/q}h^2 v_n^2} + \sqrt{p}(\delta_n + h^s) \right\}. \tag{3.5}$$

On samples with a singular local design or a nonpositive fitted variance, set T_n, S_n^*, G_n^* to zero and the bands to the empty set. The oracle statistic T_n° keeps its earlier convention. The proofs include the probability of these exceptional samples. The first two terms in e_n control empirical covariance fluctuations;

This bound uses the same observations for estimation and calibration.

Theorem 3 (Bootstrap approximation)**.** *Under Assumptions 1–5, if* $e_n \to 0$*, then*

$$\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - \mathbb{P}(S_{G,n} \le z)| = O_P(\sqrt{e_n}\, L_n),$$

$$\sup_z |\mathbb{P}(T_n \le z) - \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n)| = O_P(\gamma_n + \sqrt{e_n}\, L_n).$$

The bounds are consistent if $\gamma_n \to 0$ and $e_n L_n^2 \to 0$. These conditions specify the admissible joint growth of dimension and sampling frequency, together with the bandwidth; they do not treat p as a constant.

3.3 Simultaneous confidence bands

For $v \in (0,1)$, define the conditional quantile

$$\widehat{q}_v = \inf\{z : \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) \ge v\}. \tag{3.6}$$

At confidence level $1-\alpha$, the bands are

$$\mathcal{C}_{1-\alpha,k}(t) = \left[\widehat{\beta}_k(t) \;\pm\; \frac{\widehat{q}_{1-\alpha}\widehat{\sigma}_k(t)}{\sqrt{n}}\right], \qquad t \in \mathcal{T}, \quad k = 1, \dots, p.$$

The common critical value calibrates coverage jointly over time and coefficient functions.

Theorem 4 (Simultaneous coverage)**.** *Under Assumptions 1–5, suppose* $\gamma_n \to 0$ *and* $e_n L_n^2 \to 0$*. For every fixed* $\alpha \in (0,1)$*,*

$$\left|\mathbb{P}\{\beta_k(t) \in \mathcal{C}_{1-\alpha,k}(t) \text{ for all } t \in \mathcal{T},\ k \le p\} - (1-\alpha)\right| \le C\{\gamma_n + (e_n L_n^2)^{1/3}\}.$$

The bands use the uncorrected local polynomial estimate, so the bandwidth must make smoothing bias negligible for simultaneous inference. In particular, $\gamma_n \to 0$ includes the undersmoothing requirement

$$\frac{\sqrt{npL_n}\, h^{r+1}}{v_n} \to 0. \tag{3.7}$$

The uniform bias bound is $O(\sqrt{p}\, h^{r+1})$, and the coordinate standard errors have order $v_n/\sqrt{n}$. The additional factor $\sqrt{L_n}$ controls the effect of this bias on the distribution of the joint maximum. The remaining terms of γ_n and e_n also restrict how small h may be. The multiplier bootstrap calibrates stochastic fluctuations; the displayed bandwidth condition controls smoothing bias.

The coverage bound follows from a probability bound for covariance error and a bracket for the random critical value. In practice, $\widehat{q}_{1-\alpha}$ is approximated by the empirical quantile of independent multiplier draws. Section S3.3 describes simulation and grid error. The supplementary results also sharpen the rates under comparable counts.

3.4 Phase transition for inference

To display the role of repeated measurements, take p fixed and $m_i = m = m_n$ in this subsection. Then $K_n = O(1)$ and the coordinate standard errors have order $\{n^{-1} + (nmh)^{-1}\}^{1/2}$. The following specialization describes the transition through both Gaussian approximation and band width.

Corollary 1 (Gaussian approximation and band width)**.** *Under Assumptions 1–5, take fixed p, $m_i = m$, and $q > 4$. Suppose $h \geq n^{-A}$ for some fixed $A > 0$, and*

$$\begin{aligned} n \wedge (nmh) &\gtrsim n^{4/q}(\log n)^{3-4/q}, \\ \{n \wedge (nmh)\}^{3/4} h^s &\lesssim (\log n)^{5/4}. \end{aligned} \tag{3.8}$$

Then, for all sufficiently large n,

$$\begin{aligned} &\sup_z |\mathbb{P}(T_n^\circ \leq z) - \mathbb{P}(S_{G,n} \leq z)| \\ &\quad \leq C \left\{ \frac{\log^7 n}{n \wedge (nmh)} \right\}^{1/4}. \end{aligned} \tag{3.9}$$

For every fixed $\alpha \in (0,1)$*, the bands have simultaneous coverage tending to* $1-\alpha$*. Their half-widths* $w_{n,k}(t) = \widehat{q}_{1-\alpha}\widehat{\sigma}_k(t)/\sqrt{n}$ *satisfy, with probability tending to one,*

$$c\left\{\frac{1}{\sqrt{n}} + \sqrt{\frac{\log(1/h)}{nmh}}\right\} \le w_{n,k}(t) \le C\left\{\frac{1}{\sqrt{n}} + \sqrt{\frac{\log(1/h)}{nmh}}\right\}, \qquad t \in \mathcal{T}, \qquad k \le p. \tag{3.10}$$

The first balance in (3.8) makes the finite-moment term no larger than the Gaussian approximation term; the second places smoothing bias below the same bound. These are sufficient simplifications of the general rates. They hold, for example, for $q = 8$, $r \ge 1$, and $h = n^{-2/5}$ for every $m \ge 1$. Section S3.1 gives the derivation and verifies the coverage conditions.

From the Gaussian approximation perspective, $mh \le 1$ gives the bound $\{\log^7 n/(nmh)\}^{1/4}$, whereas $mh \ge 1$ gives $(\log^7 n/n)^{1/4}$. Thus the effective sample size increases with the number nmh of observations in local windows and reaches the order of the number n of independent subjects at $mh \asymp 1$. The reference Gaussian law retains the corresponding covariance throughout this transition.

Band width gives a complementary perspective. When $mh \ll \log(1/h)$, the half-width has order $\{\log(1/h)/(nmh)\}^{1/2}$; when $mh \gtrsim \log(1/h)$, it has order $n^{-1/2}$. The additional logarithm is the cost of simultaneous coverage

over time. Consequently, the crossover for band width occurs later than the crossover for coordinate variance.

Repeated measurements reduce the uncertainty from observing a trajectory at finitely many times. They continue to share the same subject-specific functional error, however, so increasing their number cannot remove variation between subjects. Once the local sampling contribution is small, further visits mainly resolve an already observed trajectory and yield diminishing improvements in inference. Increasing the number of independent subjects reduces the remaining uncertainty and narrows the bands beyond this plateau.

3.5 A mean-function corollary

The same subject-score argument applies to a growing collection of mean functions. Replace the random predictor vector by a scalar intercept in each of $p = p_n$ response coordinates:

$$Y_{ij,k} = \mu_k(T_{ij}) + \varepsilon_{i,k}(T_{ij}), \qquad j \le m_i, \quad k \le p = p_n.$$

The mean-zero error processes are independent and identically distributed across subjects and independent of the observation times; their coordinates may be dependent. The counts m_i are deterministic positive integers, and all observation times are mutually independent with common density f.

Keep the sampling-density, kernel, and bandwidth conditions from Section 2.2, and suppose the μ_k belong to a common bounded $C^{r,1}(\mathcal{T})$ class. Retain polynomial moment assumptions: for fixed $q \geq 4$,

$$\begin{aligned} \max_{k\leq p} \mathbb{E}\{\sup_t |\varepsilon_{i,k}(t)|^q\} \leq C, \\ \left\| 1 + \sup_t \max_{k\leq p} |\varepsilon_{i,k}(t)| \right\|_{L^q} \leq K_n, \qquad 1 \leq K_n \leq Cp^{1/q}. \end{aligned} \tag{3.11}$$

The covariance entries $C_{kl}(u,t) = \mathbb{E}\{\varepsilon_{i,k}(u)\varepsilon_{i,l}(t)\}$ are jointly Lipschitz with one common constant, and $\inf_{t,k} C_{kk}(t,t) \geq c > 0$. These are coordinatewise conditions; the full covariance may be singular.

Fit each mean by the equal-subject local polynomial criterion (2.2), using scalar predictor one and response $Y_{ij,k}$. All p fits share the same $(r+1)\times(r+1)$ matrix

$$\widehat{H}(t) = \frac{1}{n}\sum_i \frac{1}{m_i}\sum_j K_h(T_{ij}-t)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)^{\intercal}.$$

Let $\widehat{\mu}_k$ denote the fitted intercept, and put $\widehat{\varepsilon}_{ij,k} = Y_{ij,k} - \widehat{\mu}_k(T_{ij})$. With $\mathbf{e}_0 = (1,0,\ldots,0)^{\intercal} \in \mathbb{R}^{r+1}$, the fitted subject contributions are

$$\widehat{\Psi}_{ik}(t) = \mathbf{e}_0^{\intercal}\widehat{H}(t)^{-1}\frac{1}{m_i}\sum_j K_h(T_{ij}-t)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)\widehat{\varepsilon}_{ij,k}.$$

Their oracle versions use $H = \mathbb{E}\widehat{H}$ and the true errors. Define τ_k, $\widehat{\sigma}_k$, and the subject multiplier bootstrap as before, with one multiplier shared across all coordinates of a subject. In this subsection, $T_n^{\circ}, T_n, S_n^*, S_{G,n}$ denote the

corresponding mean-model statistics, and the bands $\mathcal{C}_{1-\alpha,k}$ are centered at $\widehat{\mu}_k$, with the same exceptional-sample conventions. The scales v_n, L_n remain those in (3.1), and δ_n uses the mean envelope K_n in (3.11).

Only the design-concentration and coordinate-mixing steps change. Define the scalar-design rate and the resulting approximation errors by

$$\begin{aligned} a_n &= \sqrt{\frac{\kappa_{1,n}\log(n/h)}{nh}} + \frac{\log(n/h)}{nh}, \\ \gamma_n^{\mu} &= \left(\frac{K_n^2 L_n^5}{nh^2 v_n^2}\right)^{1/4} + \frac{K_n L_n^{3/2-1/q}}{n^{1/2-1/q} h v_n} \\ &\quad + \left(\frac{\sqrt{nL_n}\, a_n \delta_n}{v_n}\right)^{1/2} + \frac{\sqrt{nL_n}\, h^s}{v_n}, \\ e_n^{\mu} &= \sqrt{\frac{\Lambda_n L_n}{n}} + \frac{K_n^2 L_n}{n^{1-2/q} h^2 v_n^2} + \delta_n + h^s. \end{aligned} \tag{3.12}$$

The shared design has fixed size, its inverse acts separately on each coordinate, and each residual uses only its own mean estimate. Thus a_n replaces the regression design rate ρ_n, and the polynomial factors in p from coordinate mixing and residual fitting disappear. The score rate δ_n and covariance scale Λ_n retain their definitions, with the actual number of responses p in L_n.

Corollary 2 (Joint mean inference under polynomial moments)**.** *Under the preceding conditions, suppose $a_n \to 0$. Then, uniformly in t, k,*

$$\widehat{\mu}_k(t) - \mu_k(t) = \frac{1}{n}\sum_i \Psi_{ik}(t) + R_{nk}(t),$$

$$\|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|_{\infty,\mathcal{T}} = O_P(\delta_n + h^s), \qquad \|\mathbf{R}_n\|_{\infty,\mathcal{T}} = O_P(a_n\delta_n + h^s).$$

Here $\mathbf{R}_n = (R_{n1}, \dots, R_{np})^\intercal$. *For all sufficiently large* n,

$$\sup_z |\mathbb{P}(T_n^\circ \le z) - \mathbb{P}(S_{G,n} \le z)| \le C\gamma_n^\mu. \tag{3.13}$$

If $e_n^\mu \to 0$, *the bootstrap conclusions are*

$$\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - \mathbb{P}(S_{G,n} \le z)| = O_P(\sqrt{e_n^\mu}\, L_n),$$

$$\sup_z |\mathbb{P}(T_n \le z) - \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n)| = O_P(\gamma_n^\mu + \sqrt{e_n^\mu}\, L_n).$$

If $\gamma_n^\mu \to 0$ *and* $e_n^\mu L_n^2 \to 0$, *then, for every fixed* $\alpha \in (0,1)$,

$$\begin{aligned} &|\mathbb{P}\{\mu_k(t) \in \mathcal{C}_{1-\alpha,k}(t) \text{ for all } t \in \mathcal{T},\ k \le p\} - (1-\alpha)| \\ &\quad \le C\{\gamma_n^\mu + (e_n^\mu L_n^2)^{1/3}\}. \end{aligned} \tag{3.14}$$

The proof specializes the linearization, covariance, and quantile-bracketing arguments of Theorems 1–4; details are in Section S4. The mean bands also use an uncorrected estimate, so $\gamma_n^\mu \to 0$ requires

$$\frac{\sqrt{nL_n}\, h^{r+1}}{v_n} \to 0. \tag{3.15}$$

This puts the bias below the coordinate standard error by a factor $\sqrt{L_n}$, which suffices for the joint-maximum comparison used here. With $m_i = m$, it is equivalent, up to constants, to $\{n \wedge (nmh)\}L_n h^{2r+2} \to 0$. For local linear fits this becomes $nmh^5 L_n \to 0$ when $mh \to 0$, and $nh^4 L_n \to 0$ when mh is bounded away from zero.

The only dimension dependence is through L_n and the envelope K_n. Coordinate polynomial moments give the worst-case bound $K_n \lesssim p^{1/q}$; a bounded or polylogarithmic envelope yields logarithmic dimension dependence without requiring exponential moments. In particular, with $q = 8$, $K_n = O(1)$, local linear fits, $m_i = 1$, and $h = n^{-1/4}$, $\log p = o(n^{3/20})$ gives coverage error $O(n^{-1/8} L_n^{5/6})$. Section S4.1 gives a polynomial-tailed model with a dimension-free envelope, demonstrating exponentially many mean functions under sparse sampling.

4. Numerical studies

4.1 Simulation design and competing methods

The factorial design compares nominal 95% bands with $n \in \{200, 400\}$ subjects, $m \in \{5, 20, 80\}$ observations per subject, and Gaussian, Laplace, or adjusted chi-square innovations in the functional errors. We refer to these three sampling frequencies as sparse, mild, and dense; these are finite-sample labels. Every configuration uses $R = 1{,}000$ independent datasets. We examine a single coefficient in a two-coefficient model, simultaneous coverage of all five coefficients in a five-coefficient model, and the intercept-only mean problem.

Observation times are independent Uniform$(0,1)$. Predictors and the single functional error are generated from smooth finite-basis processes. Error innovations have mean zero and variance one: $N(0,1)$, Laplace$(0,1/\sqrt{2})$, or $(\chi^2_5-5)/\sqrt{10}$. The latter two examine heavier symmetric tails and positive skewness, respectively. The standardized χ^2_5 law has skewness $\sqrt{8/5}$ and excess kurtosis $12/5$, providing a moderately skewed case. The coefficient functions include linear, oscillating, quadratic, and exponential trends. Section S5.1 gives the complete generating mechanism and targets. We use local linear fits with the Epanechnikov kernel, one common bandwidth for the estimate and residuals, and $B = 1{,}999$ Gaussian multiplier draws. Both methods in each comparison use the same 201 equally spaced evaluation points in $[0.15, 0.85]$. Coverage probability (CP) requires coverage of the entire grid and all targeted coefficients; average width (AW) averages over that grid and those coefficients. These are grid-coverage measurements.

For coefficient bands, we compare with Cao, Liu and Zhou (2018), abbreviated CLZ. Both local linear estimators use the same bandwidth; with equal counts their point estimates agree, while covariance estimation

and calibration differ. For the five-function target, CLZ uses Bonferroni-adjusted marginal bands. Section S5.2 specifies its kernel variance estimates and finite-observation Gaussian calibration. The CLZ theory concerns sparse longitudinal sampling, so its mild and dense entries assess performance beyond that regime.

For the mean problem, we compare with the B-spline bands of Cai and Hu (2025), which also accommodate varying sampling frequencies. Their cubic-spline fit uses BIC-selected knots and a subject-score covariance estimate. The design, full results, and computational details for this specialization are in Section S5.

4.2 Coverage and width

Table 1 reports joint coverage and width for every sampling frequency and all three innovation distributions. Table S1 in the supplement gives the single-coefficient comparison. At $R = 1{,}000$, the Monte Carlo standard error of CP near 0.95 is approximately 0.007. Complete replication records, configuration-specific uncertainty, and paired coverage differences are supplied with the code. All 54,000 replications of this factorial experiment completed without a singular-fit or variance failure.

Table 1: Coverage probability (CP) and average width (AW) of nominal 95% bands for all five coefficients jointly. Each row uses 1,000 datasets; $m = 5, 20, 80$ denote sparse, mild, and dense sampling.

			Proposed		CLZ (Bonferroni)	
Innovations	n	m	CP	AW	CP	AW
Gaussian	200	5	0.822	0.703	0.458	0.566
	200	20	0.906	0.545	0.042	0.288
	200	80	0.912	0.497	0.000	0.144
	400	5	0.910	0.556	0.581	0.458
	400	20	0.919	0.412	0.073	0.229
	400	80	0.940	0.362	0.000	0.115
Laplace	200	5	0.871	0.698	0.491	0.565
	200	20	0.904	0.543	0.047	0.288
	200	80	0.914	0.492	0.000	0.143
	400	5	0.904	0.550	0.618	0.454
	400	20	0.925	0.410	0.063	0.228
	400	80	0.929	0.361	0.001	0.115
Adjusted χ^2_5	200	5	0.847	0.698	0.484	0.563
	200	20	0.901	0.545	0.044	0.289
	200	80	0.910	0.494	0.000	0.144
	400	5	0.885	0.553	0.580	0.457
	400	20	0.924	0.410	0.064	0.229
	400	80	0.934	0.362	0.000	0.115

The proposed single-coefficient CP ranges from 0.908 to 0.948, compared with 0.052 to 0.773 for CLZ. For joint inference the ranges are 0.822–0.940 and 0.000–0.618, respectively. The proposed coverage is closer to 0.95 in all 36 coefficient-model configurations. Comparing $m = 5$ with $m = 80$ at each fixed target, subject sample size, and error law, both CLZ width and coverage decrease in all twelve comparisons.

For the proposed joint bands, CP at $n = 200$ ranges from 0.822 to 0.914. At $n = 400$ the corresponding range is 0.885–0.940. These ranges make the remaining finite-sample coverage error explicit. Both Gaussian and non-Gaussian results are reported without selecting configurations by outcome.

For the mean comparison in Table S2, proposed CP is 0.919–0.954, while Cai–Hu CP is 0.932–0.958. Cai–Hu bands are narrower in all 18 mean configurations. Coverage and width should be interpreted together under the specified tuning parameters.

For Gaussian joint bands, increasing m from 20 to 80 changes CP from 0.906 to 0.912 at $n = 200$, and from 0.919 to 0.940 at $n = 400$. The two CP differences have Monte Carlo standard errors 0.013 and 0.011, respectively. Average widths decrease from 0.545 to 0.497 and from 0.412 to 0.362. The smoothing bandwidth is unchanged across m at each fixed n under the

prespecified rule. Coverage reflects calibration error and smoothing bias as well as variance, so CP and width need not change in parallel.

4.3 Multivariate functional means

We examine Corollary 2 with $n = 200, 400$ subjects, $m = 5, 20, 80$ observations per subject, and $p = 50, 100$ mean functions. The functional errors combine common and coordinate-specific innovations from the same three distributions as above, with cross-coordinate correlation $1/2$. Each $(n, m, \text{innovation law})$ uses 1,000 independent datasets and 999 subject multiplier draws. The two dimensions are nested coordinate prefixes of the same datasets and use the same multipliers. All 36 configurations are evaluated under this fixed design. Coverage requires all coordinates at all 51 equally spaced points on $[0, 1]$, including endpoints. The shared scalar design permits computation in response blocks while retaining a single joint bootstrap critical value. Section S5.5 specifies the curves, random streams, bandwidth, implementation checks, and timing convention.

Table 2 reports joint grid coverage and width for every configuration. Monte Carlo uncertainty and computation times are retained with the replication records. Across the fixed design, CP ranges from 0.782 to 0.947. At $n = 400$, $m = 80$, and $p = 100$, CP is 0.935, 0.931, and 0.928 for Gaussian,

Table 2: Joint grid coverage (CP) and average width (AW) for multivariate functional means with $n = 200, 400$, $m = 5, 20, 80$, and $p = 50, 100$. All configurations are reported. Each cell uses 1000 datasets and 999 subject multipliers on 51 points spanning $[0, 1]$.

			Gaussian		Laplace		Adjusted χ^2_5	
n	m	p	CP	AW	CP	AW	CP	AW
200	5	50	0.860	0.927	0.878	0.927	0.802	0.926
200	5	100	0.842	0.961	0.855	0.961	0.782	0.959
200	20	50	0.907	0.692	0.920	0.693	0.883	0.690
200	20	100	0.904	0.719	0.920	0.719	0.877	0.717
200	80	50	0.911	0.605	0.929	0.603	0.913	0.604
200	80	100	0.915	0.630	0.927	0.628	0.903	0.629
400	5	50	0.924	0.716	0.925	0.715	0.865	0.714
400	5	100	0.909	0.741	0.908	0.740	0.848	0.740
400	20	50	0.933	0.515	0.930	0.514	0.932	0.515
400	20	100	0.925	0.534	0.921	0.534	0.921	0.534
400	80	50	0.947	0.439	0.942	0.439	0.922	0.439
400	80	100	0.935	0.457	0.931	0.457	0.928	0.457

Laplace, and adjusted chi-square innovations, respectively. Joint undercoverage remains in some configurations. These results concern the stated finite grid and prespecified bandwidth.

4.4 Mayo PBC sequential biomarker data

We analyze the Mayo Clinic PBC sequential data available as `pbcseq` in the R package `survival` (Survival package authors, 2026), originating from the D-penicillamine trial (Dickson et al., 1985). The full file contains 1,945 visits from 312 patients. Restricting follow-up to 3,000 days and retaining complete observations with positive bilirubin and alkaline phosphatase leaves 1,741 visits from all 312 patients for the regression. Counts range from 1 to 10, with harmonic mean 3.489. Section S5 shows the observation schedule, the separate mean analysis, and preprocessing details.

The protocol included scheduled visits, with additional visits potentially prompted by worsening health; laboratory measurements can also be missing (Survival package authors, 2026). Consequently, independent observation times and noninformative follow-up are not established. We interpret the fitted bands as an exploratory application under a working sampling model and describe observed-visit associations.

Let $t = \text{day}/3000$, let D_i indicate D-penicillamine, and standardize edema, albumin, prothrombin time, and log alkaline phosphatase by their observation-level means and standard deviations, obtaining E, A, P, L.

The regression model is

$$\begin{aligned}\log(\text{bili}_{ij}) = \beta_1(t_{ij}) + \beta_2(t_{ij})D_i + \beta_3(t_{ij})E_{ij} \\ + \beta_4(t_{ij})A_{ij} + \beta_5(t_{ij})P_{ij} + \beta_6(t_{ij})L_{ij} + \varepsilon_i(t_{ij}).\end{aligned} \tag{4.1}$$

For model (4.1), equal-subject weights are recomputed after filtering. The common bandwidth is 0.09510, about 285 days. Calibration uses 9,999 multipliers and is joint over the six coefficients and 201 points from day 450 to day 2,550. The resulting critical value is 3.437; at least 95 patients contribute to each evaluation window.

In Figure 1, the treatment band contains zero throughout the displayed interval, while the log-ALP band remains above zero. Edema and prothrombin time have positive exclusions on portions of follow-up; albumin has short negative exclusions. Bandwidth factors 0.8 and 1.2 and a shorter 2,000-day horizon preserve the treatment and log-ALP descriptions but change the shorter exclusions. Full sensitivity results are in Section S5. These are descriptive adjusted associations: conditioning on contemporaneous biomarkers does not identify a randomized treatment effect, and a band containing zero does not establish equivalence.

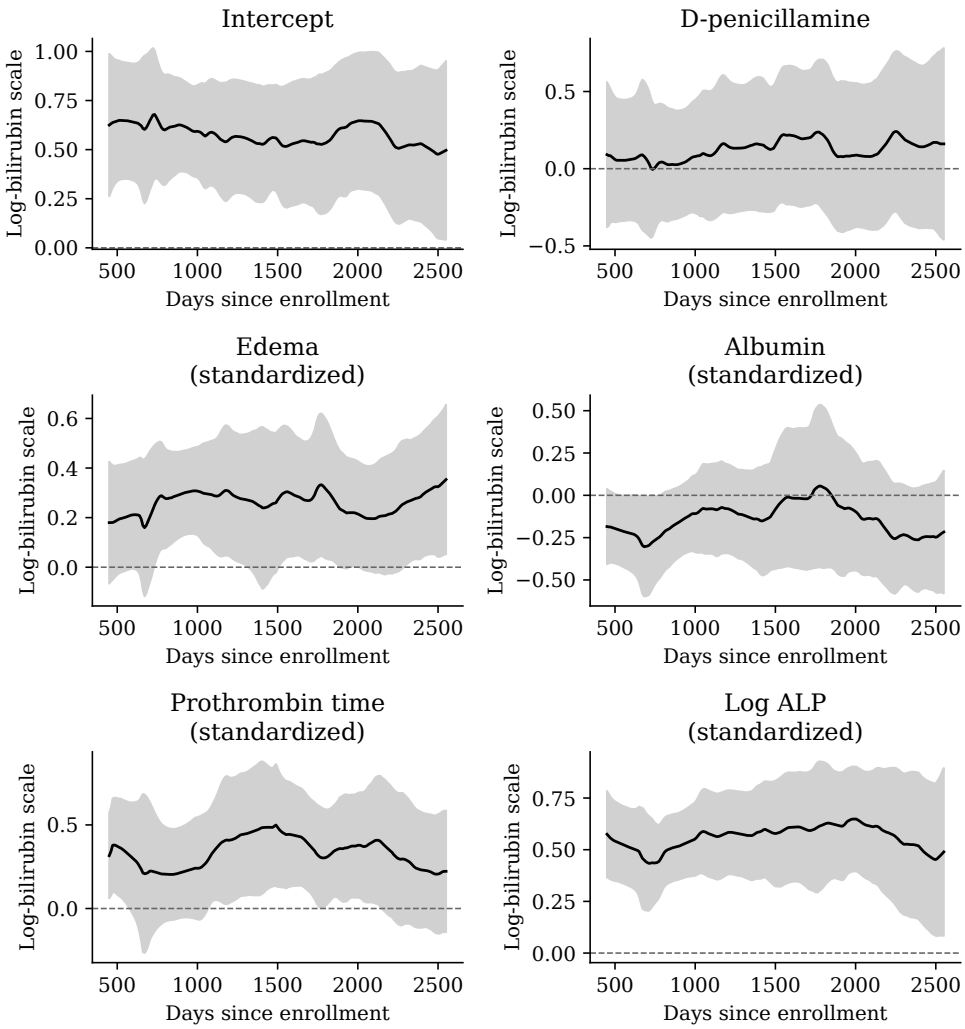


Figure 1: PBC coefficient estimates and nominal joint 95% bands under the working model. Independent visit times and noninformative follow-up are not established for these data.

5. Discussion

Subject contributions link local polynomial estimates to Gaussian and bootstrap approximation without requiring individual trajectory reconstruction. The bounds quantify the effects of sampling frequency, dimension, smoothing bias, and estimation of the design and residuals. Their explicit dimension dependence permits growing VCMs; the coordinatewise mean extension has no growing regression matrix to invert. Its dimension dependence enters through a logarithm and the maximum-coordinate moment envelope. Polynomial-tailed models with a bounded envelope allow exponentially many functions under sparse sampling.

The explicit powers of p in our bounds follow worst-case inequalities under the stated assumptions. They account for mixing coordinate scores through the inverse local design and accumulating coefficient errors in fitted residuals. Designs with diagonal second-moment matrices or fixed-size groups of correlated predictors can reduce these dimension factors when the inverse-design absolute row sums remain uniformly bounded. The stated dimension restrictions are therefore sufficient conditions and need not be sharp.

The results assume error-free predictors and observation times independent of the trajectories. They require a bandwidth satisfying the displayed

approximation and bias conditions. The numerical comparisons show that moderate-sample joint undercoverage can remain. Extensions to informative visits, measurement error in predictors, and justified finite-sample corrections are therefore useful directions for further work.

Supplementary Material

The supplementary document contains proofs, sampling-frequency and bandwidth results, the high-dimensional mean extension, and additional numerical and data-processing details. Reproduction code and complete simulation records accompany the manuscript.

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